ME-351 Feedback Control Systems Notes

The goal of a feedback control system is to *influence* a dynamical system such that it behaves in a *desired manner*, despite the presence of disturbances and uncertainties, within limits imposed by physics and hardware constraints.

We will begin with a review from ESC-251 Systems Engineering on how to develop mathematical models of dynamical systems and analyze their behavior before moving onto building intuition for how feedback control can be used to influence system behavior.

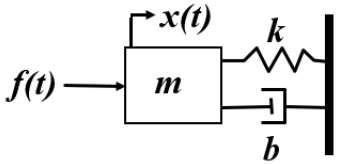
Modeling Dynamical Systems

Equations of Motion

Equations of motion describes the time evolution of a system's state variables. Where a state variable can be defined as the set of variables that completely describes the system at a given time and is able to determine the future behavior of the system.

The equations of motion themselves are derived from fundamental physical principles such as Newton's second law, conservation of energy (energy methods), and Kirchhoff's laws (electrical systems).

For example: in a simple mass-spring-damper system where $f(t)$ is some input force:



$$\sum F = m\ddot{x} \quad \Longrightarrow \quad m\ddot{x} + b\dot{x} + kx = f(t)$$

The state variables can be defined as position x and velocity $\dot{x}$.

Generally, linear time-invariant (LTI) systems can be represented by an nth order linear differential equation of the form:

$$a_n y^{(n)} + a_{n-1} y^{(n-1)} + \cdots + a_1 \dot{y} + a_0 y = b_m u^{(m)} + b_{m-1} u^{(m-1)} + \cdots + b_1 \dot{u} + b_0 u$$

Where $u(t)$ is the input and $y(t)$ is the output of the system.

A LTI system is linear if it satisfies the superposition principle, and time-invariant if its coefficients do not change with time.

For example:

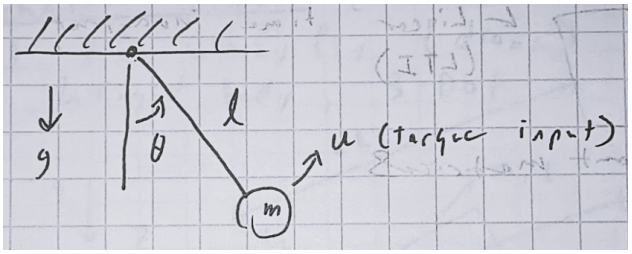
- $\ddot{x} + 3t\dot{x} + 2x = 0$ is linear but not time-invariant
- $\ddot{x} + 3\dot{x} + 2x = 0$ is linear and time-invariant
- $\ddot{x} + 3\dot{x} + 2x^2 = 0$ is nonlinear and time-invariant

Linear and Nonlinear Systems

As mentioned before, linear systems obey superposition. This is especially useful when analyzing the response of a system subjected to multiple inputs. In that case, the response of the system to each input can be analyzed separately and then summed to find the total response.

Nonlinear systems involve nonlinear relationships among their state variables and inputs, such as exponentials, products, or transcendental functions, resulting in a violation of the superposition principle.

For example: the string-mass pendulum is a nonlinear system



Summing torques about the pivot point gives:

$$\sum \tau = I\ddot{\theta} \quad \Longrightarrow \quad m\ell^2\ddot{\theta} = -mg\ell\sin(\theta) + u(t)$$

Since there is a sinusoidal relationship to the state variable θ, this system is nonlinear. However, we will see in a later section that we can linearize this system about an equilibrium.

State Space Representation

(From ESC-251 Systems Engineering) We are able to rewrite the equations of motion of a system in the time domain into a matrix form using state variables. This is useful because it allows us to write an nth order system as a system of n first order equations.

The state space representation of a system is given by the following form:

$$\underline{\dot{x}} = [A]\underline{x} + [B]\underline{u}$$
$$\underline{y} = [C]\underline{x} + [D]\underline{u}$$

Where:

- $\underline{x}$ = state vector
- $\underline{u}$ = input vector
- $\underline{y}$ = output vector
- $[A]$ = state matrix
- $[B]$ = input matrix
- $[C]$ = output matrix
- $[D]$ = direct transition matrix (output related to input)

The state vector is a column vector that contains the state variables of the system.

Starting from the equations of motion, define state variables such that we can rewrite the equations of motion as first order differential equations. In other words, "kill the dots" (get rid of higher order derivatives) by introducing a new state variable for each derivative.

For example: recall the equation of motion for a mass-spring-damper system:

$$m\ddot{x} + b\dot{x} + kx = f(t)$$

Define the state variables as:

$$x = x, \quad v = \dot{x}$$

Then we can rewrite the equation of motion as a system of first order differential equations:

$$m\dot{v} + bv + kx = f(t)$$

$$\begin{aligned} \dot{x} &= v \\ \dot{v} &= -\frac{k}{m}x - \frac{b}{m}v + \frac{1}{m}f(t) \end{aligned}$$

In state space form, we can write:

$$\underbrace{\begin{bmatrix} \dot{x} \\ \dot{v} \end{bmatrix}}_{\underline{\dot{x}}} = \underbrace{\begin{bmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{b}{m} \end{bmatrix}}_{[A]} \underbrace{\begin{bmatrix} x \\ v \end{bmatrix}}_{\underline{x}} + \underbrace{\begin{bmatrix} 0 \\ \frac{1}{m} \end{bmatrix}}_{[B]} \underbrace{f(t)}_{\underline{u}}$$

$$\underbrace{y}_{\underline{y}} = \underbrace{\begin{bmatrix} 1 & 0 \end{bmatrix}}_{[C]} \underbrace{\begin{bmatrix} x \\ v \end{bmatrix}}_{\underline{x}} + \underbrace{0}_{[D]} \underbrace{f(t)}_{\underline{u}}$$

Check that the matrix multiplication (dot product between rows and columns) correctly reproduces the rewritten equations of motion and that the dimensions are consistent.

The output matrix is often either a matrix that selects certain state variables as outputs, or a matrix that operates on the state variables to produce the output (ie. momentum is a

velocity state variable multiplied by mass). The direct transmission matrix is often zero, but it can be non-zero if the output is directly related to the input (ie. if we want to output/read the input voltage in an electrical system).

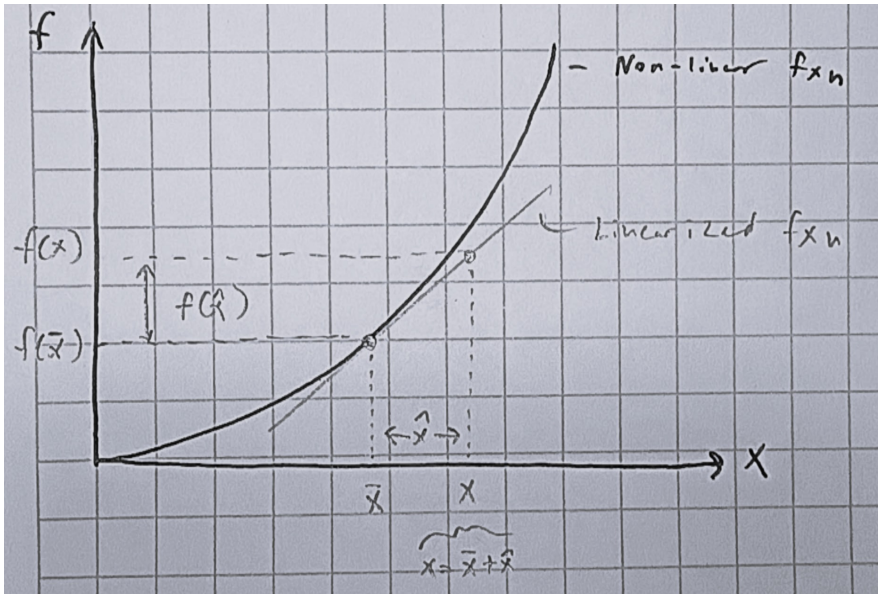
Note that the state space representation is **not unique** since we can define state variables differently (e.g., shifting, rescaling state variables) and still describe the same system.

Linearization of Nonlinear Systems

The intuition for linearization is the as from ESC-251 Systems Engineering: Generally, we do this by approximating the *local* behavior of a non-linear system around an equilibrium point (operating point) using the first order Taylor expansion.

Defining the operating point as $\bar{x}$ and the incremental variable as $\hat{x}$, we can see graphically how the linearization process works:

$$\boxed{x = \bar{x} + \hat{x}, \quad f(\hat{x}) = f(x) - f(\bar{x})}$$



Mathematically, approximating the non-linear function $f(x)$ around the operating point $\bar{x}$ gives us a tangent line that approximates the behavior of the system near $\bar{x}$:

$$k = \left.\frac{df}{dx}\right|_{x=\bar{x}}$$

$$f(x) = f(\bar{x}) + k(x - \bar{x})$$

Using the definition of $\hat{x}$, we can rewrite the linearized function in terms of the incremental variable $\hat{x}$:

$$f(x) = f(\bar{x}) + k\hat{x} \quad \Longrightarrow \quad \boxed{f(\hat{x}) = k\hat{x}}$$

Now we are able to apply our techniques for linear systems to anaylze the behavior of the system under $\hat{x}$, however we must remember that the *total* response of the system is given by

$$f(x) = f(\bar{x}) + f(\hat{x}), \quad x = \bar{x} + \hat{x}$$

Effectively, this linearization technique is like a transformation from the original $(x, f(x))$ coordinate system to a new $(\hat{x}, f(\hat{x}))$ coordinate system that is centered around the operating point $(\bar{x}, f(\bar{x}))$. So we need to remember to transform the results back to the original coordinate system when necessary.

Note: The operating point is defined as the equilibrium point of the system since the further away from the operating point we are, the worse the linear approximation will be.

Linearization Procedure

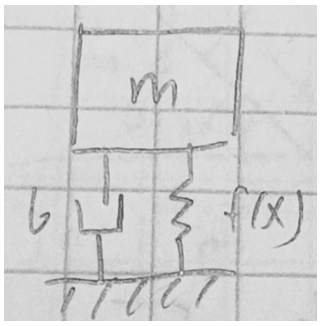
I. **Define operating point & incremental variables:**
Let $f(x)$ be a non-linear function of x, let $\bar{x}$ be the nominal operating point and let $\hat{x} = x - \bar{x}$ be the deviation from the operating point. Linearize $f(x)$ using Taylor series or graphical method.

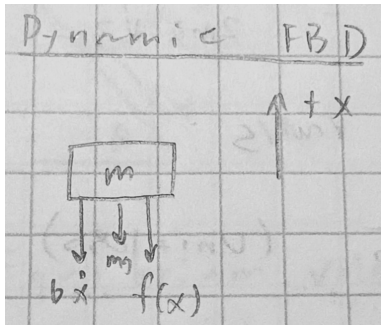
II. **Find operating point:**
To find the operating point, find the "equilibrium" solution of the system when $\bar{x} =$ (constant) by setting $\dot{x} = \ddot{x} = 0$.

III. **Replace with linearized terms:**
Replace the linearized $f(x)$ and $x = \hat{x} + \bar{x}$ (and $u = \hat{u} + \bar{u}$ if needed) in the non-linear differential equation.

The linearized equation is in terms of $\hat{x}$ (and $\hat{u}$ if it exists). This linearized equation describes the actual non-linear system close to the operating point. The further from the operating point, the less accurate the description is.

For example: **Non-linear spring-mass-damper system**
Where the spring force is non-linear and given by $f(x) = ax^3$:





The equation of motion is given by:

$$\sum F = m\ddot{x} \quad \Longrightarrow \quad m\ddot{x} + b\dot{x} + ax^3 = -mg$$

Follow the linearization procedure to find the linearized equation of motion:

I. Define variables and linearize $f(x)$:

$$x = \bar{x} + \hat{x}$$

$$\begin{aligned} f(x) &= f(\bar{x}) + \left.\frac{df}{dx}\right|_{x=\bar{x}} (x - \bar{x}) \\ &= a\bar{x}^3 + 3a\bar{x}^2(\hat{x}) \end{aligned}$$

II. Find the operating (equilibrium) point by setting $\dot{x} = \ddot{x} = 0$:

$$a\bar{x}^3 = -mg$$

$$\implies \bar{x} = \sqrt[3]{-\frac{mg}{a}}$$

III. Linearize the equation of motion:

$$m\frac{d^2(\bar{x} + \hat{x})}{dt^2} + b\frac{d(\bar{x} + \hat{x})}{dt} + (a\bar{x}^3 + 3a\bar{x}^2\hat{x}) = -mg$$

Since $\bar{x}$ is constant, $\dot{\bar{x}} = \ddot{\bar{x}} = 0$:

$$m\frac{d^2(\hat{x})}{dt^2} + b\frac{d(\hat{x})}{dt} + (a\bar{x}^3 + 3a\bar{x}^2\hat{x}) = -mg$$

Substituting in the equilibrium condition $a\bar{x}^3 = -mg$:

$$m\frac{d^2(\hat{x})}{dt^2} + b\frac{d(\hat{x})}{dt} + 3a\bar{x}^2\hat{x} = 0$$

Now we have a linearized equation of motion in terms of the incremental variable $\hat{x}$ that describes the behavior of the non-linear system near the operating point $\bar{x}$.

Jacobian Linearization Method

The Jacobian linearization method is a systematic way to linearize a non-linear system with non-linear multiple state variables.

Definition: A Jacobian matrix is a matrix containing all first-order partial derivatives of a vector-valued function with respect to its input variables:

$$\underline{f}(\underline{x}) = \begin{bmatrix} f_1(x_1, \ldots, x_n) \\ \vdots \\ f_m(x_1, \ldots, x_n) \end{bmatrix}, \qquad \underline{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$$

$$\frac{\partial \underline{f}}{\partial \underline{x}} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1} & \cdots & \frac{\partial f_m}{\partial x_n} \end{bmatrix}$$

Consider a general nonlinear system in state-space form:

$$\boxed{\begin{aligned} \dot{\underline{x}} &= \underline{f}(\underline{x}, u) \\ \underline{y} &= \underline{h}(\underline{x}, u) \end{aligned}}$$

where $\underline{f}$ is a vector-valued function describing the time evolution of the state variables, and $\underline{h}$ is a function relating the state variables and inputs to the system outputs. Unlike an LTI system, these functions may contain nonlinear relationships.

Example: Pendulum System
Recall the pendulum system with equation of motion:

$$m\ell^2\ddot{\theta} = -mg\ell\sin(\theta) + u(t)$$

Let $\omega = \dot{\theta}$ and define the state vector:

$$\underline{x} = \begin{pmatrix} \theta \\ \omega \end{pmatrix}$$

Then we can rewrite the system in state-space form:

$$\underbrace{\begin{pmatrix} \dot{\theta} \\ \dot{\omega} \end{pmatrix}}_{\dot{\underline{x}}} = \underbrace{\begin{pmatrix} \omega \\ -\frac{g}{\ell}\sin(\theta) + \frac{1}{m\ell^2}u(t) \end{pmatrix}}_{\underline{f}(\underline{x},u)}$$

With an output of $y = \theta$:

$$\underbrace{\theta}_{y} = \underbrace{\begin{pmatrix} 1 & 0 \end{pmatrix}\underline{x}}_{h(\underline{x},u)}$$

Notice that the state equation is nonlinear due to the $\sin(\theta)$ term, while the output equation is linear.

Linearization Procedure

I. Define operating point & incremental variables:
Let $\dot{x} = \underline{f}(\underline{x}, \underline{u})$ be a nonlinear system, and $(\bar{\underline{x}}, \bar{\underline{u}})$ be the operating point. Define incremental variables:

$$\hat{\underline{x}} = \underline{x} - \bar{\underline{x}}, \quad \hat{\underline{u}} = \underline{u} - \bar{\underline{u}}$$

Linearize $\underline{f}(\underline{x}, \underline{u})$ using the first order Taylor expansion, where the Jacobian matrices are evaluated at the operating point $(\bar{\underline{x}}, \bar{\underline{u}})$:

$$\underline{f}(\underline{x}, \underline{u}) \approx \underline{f}(\bar{\underline{x}}, \bar{\underline{u}}) + \left.\frac{\partial \underline{f}}{\partial \underline{x}}\right|_{(\bar{\underline{x}},\bar{\underline{u}})} \hat{\underline{x}} + \left.\frac{\partial \underline{f}}{\partial u}\right|_{(\bar{\underline{x}},\bar{\underline{u}})} \hat{\underline{u}}$$

II. Find operating point:
To find the operating point, solve for the equilibrium of the system when $\underline{\bar{x}}$ is constant by setting $\dot{\underline{\bar{x}}} = \underline{f}(\underline{x}, \underline{u}) = 0$.

III. Replace with linearized terms:
Substitute the linearized $\underline{f}(\underline{x}, \underline{u})$ and incremental variables into the nonlinear state space equations. Applying the equilibrium condition $\underline{f}(\underline{\bar{x}}, \underline{\bar{u}}) = 0$ gives the linearized state space equations:

$$\dot{\underline{\hat{x}}} = A\underline{\hat{x}} + B\underline{\hat{u}}$$

Where

$$A = \left.\frac{\partial \underline{f}}{\partial \underline{x}}\right|_{(\underline{\bar{x}}, \underline{\bar{u}})}, \quad B = \left.\frac{\partial \underline{f}}{\partial \underline{u}}\right|_{(\underline{\bar{x}}, \underline{\bar{u}})}$$

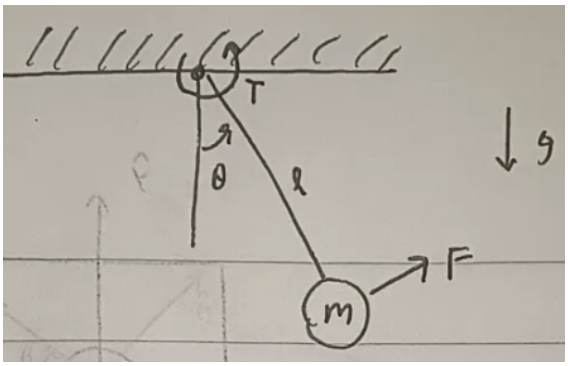
We can similarly linearize the output equation $\underline{y} = \underline{h}(\underline{x}, \underline{u})$ with respect to the incremental variable $\underline{\hat{y}} = \underline{y} - \underline{\bar{y}}$:

$$\underline{\hat{y}} = C\underline{\hat{x}} + D\underline{\hat{u}}$$

Where

$$C = \left.\frac{\partial \underline{h}}{\partial \underline{x}}\right|_{(\underline{\bar{x}}, \underline{\bar{u}})}, \quad D = \left.\frac{\partial \underline{h}}{\partial \underline{u}}\right|_{(\underline{\bar{x}}, \underline{\bar{u}})}$$

Example: Pendulum System on Steroids
Consider a similar pendulum system but now with a nonlinear torque input $T = k_\tau \sin(u_1)$ and force input $F = k_F \sin(u_2)$, where u_1 and u_2 are independent inputs:



The equation of motion is given by:

$$\sum \tau = I\ddot{\theta} \quad \Longrightarrow \quad m\ell\ddot{\theta} = -mg\ell \sin\theta + k_\tau \sin(u_1) + \ell k_F \sin(u_2)$$

Define $\omega = \dot{\theta}$ and the state vector:

$$\underline{x} = \begin{pmatrix} \theta \\ \omega \end{pmatrix}, \quad \underline{u} = \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}$$

Then, the nonliner state-space representation of the system is:

$$\dot{\underline{x}} = \begin{pmatrix} \omega \\ -\frac{g}{\ell}\sin\theta + \frac{k_\tau}{m\ell^2}\sin(u_1) + \frac{k_F}{m\ell}\sin(u_2) \end{pmatrix} \quad \Longleftrightarrow \quad \dot{\underline{x}} = \underline{f}(\underline{x}, \underline{u})$$

With an output of $y = \theta$:

$$y = \begin{pmatrix} 1 & 0 \end{pmatrix} \underline{x} \quad \Longleftrightarrow \quad y = \underline{h}(\underline{x}, \underline{u})$$

Linearization procedure:

I. Define operating point & incremental variables:
Let $(\underline{\bar{x}}, \underline{\bar{u}})$ be the operating point, define incremental variables:

$$\underline{x} = \underline{\bar{x}} + \underline{\hat{x}}, \quad \underline{u} = \underline{\bar{u}} + \underline{\hat{u}}, \quad \underline{y} = \underline{\bar{y}} + \underline{\hat{y}}$$

Linearize the nonlinear state equation using the first order Taylor expansion with Jacobian matrices evaluated at the operating point:

$$\underline{f}(\underline{x}, \underline{u}) \approx \underline{f}(\underline{\bar{x}}, \underline{\bar{u}}) + \left.\frac{\partial \underline{f}}{\partial \underline{x}}\right|_{(\underline{\bar{x}}, \underline{\bar{u}})} \underline{\hat{x}} + \left.\frac{\partial \underline{f}}{\partial \underline{u}}\right|_{(\underline{\bar{x}}, \underline{\bar{u}})} \underline{\hat{u}}$$

Calculate the Jacobian matrices:

- State Jacobian $[A]$:

$$[A] = \left.\begin{pmatrix} \frac{\partial f_1}{\partial \theta} & \frac{\partial f_1}{\partial \omega} \\ \frac{\partial f_2}{\partial \theta} & \frac{\partial f_2}{\partial \omega} \end{pmatrix}\right|_{(\underline{\bar{x}}, \underline{\bar{u}})} = \begin{pmatrix} 0 & 1 \\ -\frac{g}{\ell} \cos \bar{\theta} & 0 \end{pmatrix}$$

- Input Jacobian $[B]$:

$$[B] = \left.\begin{pmatrix} \dfrac{\partial f_1}{\partial u_1} & \dfrac{\partial f_1}{\partial u_2} \\ \dfrac{\partial f_2}{\partial u_1} & \dfrac{\partial f_2}{\partial u_2} \end{pmatrix}\right|_{(\underline{\bar{x}}, \underline{\bar{u}})} = \begin{pmatrix} 0 & 0 \\ \dfrac{k_\tau}{m\ell^2} \cos \bar{u}_1 & \dfrac{k_F}{m\ell} \cos \bar{u}_2 \end{pmatrix}$$

II. Find operating (equilibrium) point:
At equilibrium, $\dot{\underline{\bar{x}}} = \underline{f}(\underline{\bar{x}}, \underline{\bar{u}}) = 0$. Thus,

$$\dot{\underline{x}}\big|_{(\underline{\bar{x}}, \underline{\bar{u}})} = \begin{pmatrix} \bar{\omega} \\ -\frac{g}{\ell} \sin \bar{\theta} + \frac{k_\tau}{m\ell^2} \sin(\bar{u_1}) + \frac{k_F}{m\ell} \sin(\bar{u_2}) \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Solving these conditions would fully determine the operating point $(\underline{\bar{x}}, \underline{\bar{u}})$. However, we can leave it symbolic for now.

III. Replace with linearized terms and apply equilibrium condition:

$$\left(\dot{\underline{\bar{x}}} + \dot{\underline{\hat{x}}}\right) \approx \underline{f}(\underline{\bar{x}}, \underline{\bar{u}}) + A\underline{\hat{x}} \; + B\underline{\hat{u}}$$

Applying the equilibrium condition $\underline{f}(\underline{\bar{x}}, \underline{\bar{u}}) = 0$ gives the linearized state equation:

$$\dot{\underline{\hat{x}}} = A\underline{\hat{x}} + B\underline{\hat{u}}$$

Finally substituting the Jacobian matrices $[A]$ and $[B]$ gives:

$$\boxed{\underbrace{\begin{pmatrix}\dot{\hat{\theta}}\\ \dot{\hat{\omega}}\end{pmatrix}}_{\dot{\hat{x}}} = \underbrace{\begin{pmatrix}0 & 1\\ -\frac{g}{\ell}\cos\bar{\theta} & 0\end{pmatrix}}_{A}\underbrace{\begin{pmatrix}\hat{\theta}\\ \hat{\omega}\end{pmatrix}}_{\hat{x}} + \underbrace{\begin{pmatrix}0 & 0\\ \frac{k_\tau}{m\ell^2}\cos\bar{u}_1 & \frac{k_F}{m\ell}\cos\bar{u}_2\end{pmatrix}}_{B}\underbrace{\begin{pmatrix}\hat{u}_1\\ \hat{u}_2\end{pmatrix}}_{\hat{u}}}$$

We can linearize the output equation by calculating the remaining Jacobian matrices:

$$C = \left.\frac{\partial h}{\partial x}\right|_{(\underline{\bar{x}},\underline{\bar{u}})} = \left(\frac{\partial h}{\partial \theta} \quad \frac{\partial h}{\partial \omega}\right)\Big|_{(\underline{\bar{x}},\underline{\bar{u}})} = \begin{pmatrix}1 & 0\end{pmatrix}$$

$$D = \left.\frac{\partial h}{\partial u}\right|_{(\underline{\bar{x}},\underline{\bar{u}})} = \left(\frac{\partial h}{\partial u_1} \quad \frac{\partial h}{\partial u_2}\right)\Big|_{(\underline{\bar{x}},\underline{\bar{u}})} = \begin{pmatrix}0 & 0\end{pmatrix}$$

Thus,

$$\hat{y} = C\hat{x} + D\hat{u} = \begin{pmatrix}1 & 0\end{pmatrix}\hat{x}$$

Laplace Transforms

Alas, it is time to move from the time domain to the Laplace domain. It turns differentiation and integration into algebraic operations, which makes it easier to analyze the system behavior.

Definition: The Laplace transform of a function $f(t)$ is defined as:

$$F(s) = \mathcal{L}\{f(t)\} = \int_{0^-}^{\infty} f(t)e^{-st}dt$$

Where s is a complex variable.

Note: We use 0^- as the lower limit of integration to account for any impulses which may occur at $t = 0$.

With *zero initial conditions*, differentiation becomes multiplication by s in the Laplace domain:

$$\frac{d}{dt} \leftrightarrow s$$

and integration becomes division by s:

$$\int (\,)\,dt \leftrightarrow \frac{1}{s}$$

Useful Properties of Laplace Transforms

- **Linearity:**
$$\mathcal{L}\{\alpha f(t) + \beta g(t)\} = \alpha F(s) + \beta G(s)$$

- **Differentiation:**
First derivative:
$$\mathcal{L}\left\{\frac{d}{dt}f(t)\right\} = sF(s) - f(0^-)$$
Second derivative:
$$\mathcal{L}\left\{\frac{d^2}{dt^2}f(t)\right\} = s^2F(s) - sf(0^-) - f'(0^-)$$
n^th derivative:
$$\mathcal{L}\left\{\frac{d^n}{dt^n}f(t)\right\} = s^nF(s) - s^{n-1}f(0^-) - s^{n-2}f'(0^-) - \cdots - f^{(n-1)}(0^-)$$
For zero initial conditions in transfer function analysis, we get the nice simplification:
$$\mathcal{L}\left\{\frac{d^n}{dt^n}f(t)\right\} = s^nF(s)$$

- **Integration:**
$$\mathcal{L}\left\{\int_{0^-}^{t} f(\tau)d\tau\right\} = \frac{1}{s}F(s)$$

- **Time Shift:**
The unit-step function (Heaviside function) is defined as:
$$u_s(t) = \begin{cases} 0 & t < 0 \\ 1 & t \geq 0 \end{cases}$$
Thus, it is zero up to $t = 0$ and one for $t \geq 0$. The shifted unit-step function $u_s(t - T)$ is then zero up to $t = T$ and one for $t \geq T$.

The Laplace transform of a function shifted in time is a multiplication by an exponential in the Laplace domain:
$$\mathcal{L}\{f(t - T)u_s(t - T)\} = e^{-sT}F(s)$$
Note that
$$f(t - T)u_s(t - T) = \begin{cases} 0 & t < T \\ f(t - T) & t \geq T \end{cases}$$
A pure time delay thus has the transfer function $G(s) = e^{-sT}$.

- **Convolution:**
  Convolution by itself isn't very useful in its time domain form, since it is defined as:

$$(f * g)(t) = \int_0^t f(\tau)g(t-\tau)d\tau$$

$$\mathcal{L}\{f(t) * g(t)\} = F(s)G(s)$$

  This property becomes more interesting once we define $h(t)$ as the output of a linear time-invariant system subjected to a unit impulse input $\delta(t)$, assuming zero initial conditions. Thus, convolution can be used to find the output of a system subjected to any arbitrary input $u(t)$:

$$y(t) = h(t) * u(t)$$

  Intuitively, this convolution treats an arbitrary input $u(t)$ as a continuous sum of time-shifted impulses, kind of like modeling a collection of time-shifted initial-condition responses. By linearity, the total output at time t is the superposition of the individual responses produced by all impulses applied between 0 and t. (This sum is represented by the convolution integral!)

  Sound familiar? This is exactly the fundamental principle behind the transfer function representation of a system where

$$\mathcal{L}\{h(t)\} = G(s)$$

$$\implies G(s) = \frac{Y(s)}{U(s)} \quad \text{(transfer function definition)}$$

  See the "Impulse Response of a Linear System" PDF for more detailed info

Essential Laplace Transform Pairs

- An impulse input $\delta(t)$ has a Laplace transform of 1:

$$\mathcal{L}\{\delta(t)\} = 1$$

- A unit step input $u_s(t)$ has a Laplace transform of $\frac{1}{s}$:

$$\mathcal{L}\{1\} = \frac{1}{s}, \quad t \geq 0$$

- A ramp input $tu_s(t)$ has a Laplace transform of $\frac{1}{s^2}$:

$$\mathcal{L}\{t\} = \frac{1}{s^2}, \quad t \geq 0$$

- Poles of a transfer function generate exponentials in the time domain:

$$\mathcal{L}\{e^{at}\} = \frac{1}{s-a}, \quad t \geq 0$$

  (Will be covered in more detail in the upcoming sections)

Initial/Final Value Theorems

We can determine the initial and final values of a system's response by just looking at the transfer function without having to find the full time domain response.

Initial Value Theorem

$$\boxed{\lim_{t\to 0^+} f(t) = \lim_{s\to\infty} sF(s)}$$

Note that the initial value theorem applies to the right handed limit $t \to 0^+$, and does not contain an impulse that may occur at $t = 0$. Thus, the two-sided limits $t \to 0^-$ and $t \to 0^+$ do not need to be equal if there is an impulse at $t = 0$.

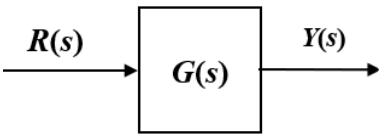
Final Value Theorem

$$\boxed{\lim_{t\to\infty} f(t) = \lim_{s\to 0} sF(s)}$$

Transfer Function Representation

Transfer functions relate input to output in the Laplace domain, *assuming zero initial conditions*. It is defined as:

$$G(s) = \frac{Y(s)}{R(s)}$$



Transfer functions *don't capture the time domain response of the system*, because the input function is undefined. However, once we define an input function (step function, impulse, etc.), we can find the time domain response by taking the inverse Laplace transform of $Y(s) = G(s)R(s)$.

Recall that transfer function analysis assumes zero initial conditions.

Note: When writing out the equations of motion (time domain), write it in the form

$$\text{(output)} = \text{(input)}$$

$$\text{(dep. variables LHS)} = \text{(input RHS)}$$

Transfer Function Matrices

Like how we can derive transfer functions in the Laplace domain from the equations of motion, we can do the same in state space representation using matrix arithmetic. Starting with:

$$\underline{\dot{x}} = [A]\underline{x} + [B]\underline{u}$$
$$\underline{y} = [C]\underline{x} + [D]\underline{u}$$

Taking the Laplace transform:

$$s\underline{X} = [A]\underline{X} + [B]\underline{U}$$
$$\underline{Y} = [C]\underline{X} + [D]\underline{U}$$

Isolate the State Vector ($\underline{X}$):

$$\underline{X} = [sI - A]^{-1}[B]\underline{U}$$

Substitute $\underline{X}$ into the Output Equation:

$$\underline{Y} = [C]\underbrace{\left([sI - A]^{-1}[B]\underline{U}\right)}_{\text{Sub for } \underline{X}} + [D]\underline{U}$$

Factor out $\underline{U}$ for the Transfer Function:

$$\boxed{\frac{\underline{Y}}{\underline{U}} = [C][sI - A]^{-1}[B] + [D]}$$

Note that this ins't super important to memorize but its good to have available as a reference.

Recall: We can invert a 2x2 matrix using the formula:

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

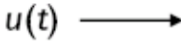
$$A^{-1} = \frac{1}{ad - bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}, \quad \text{if } ad - bc \neq 0.$$

Block Diagram Algebra

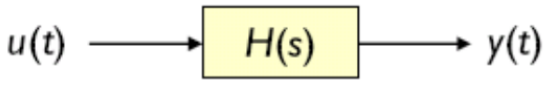
This section will mostly review block diagram elements and algebraic manipulations to simplify block diagrams.

Elements

- **Signals**: time varying quantities that represent inputs and outputs of the system. Inputs are arrows pointing into the system, while outputs are arrows pointing out of the system.

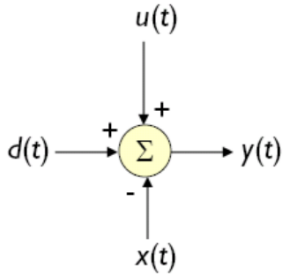


- **Systems**: represent the equations of motion that relate inputs to outputs, often represented as blocks containing the transfer function of a system component



$$Y(s) = H(s)U(s)$$

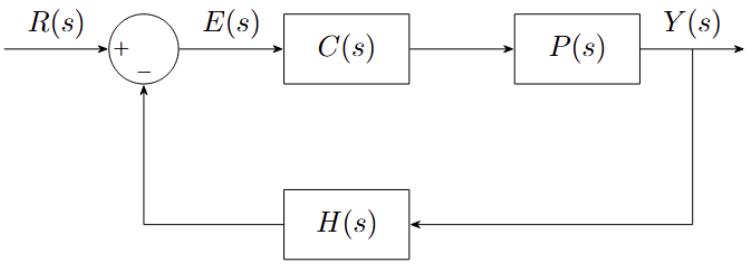
- **Summing Junctions**: adds together signals with sign indicated by the arrowheads. Signals can be added together in both the time domain and the Laplace domain.



$$y(t) = u(t) + d(t) - x(t) \implies Y(s) = U(s) + D(s) - X(s)$$

Loop Transfer Functions

Consider the following general closed-loop feedback system:



Note the following signals:

- $R(s)$: reference input
- $E(s) = R(s) - Y(s)$: error signal
- $Y(s)$: output signal

Also, $H(s)$ is the feedback transfer function where $H(s) = 1$ for unity feedback.

Some useful transfer functions to know are:

- **Feedforward Transfer Function**
  Multiply the gains of the forward path (aka forward gain)

$$\frac{Y(s)}{E(s)} = C(s)P(s)$$

- **Open Loop Transfer Function**
  Multiply the gains around the loop (aka loop gain)

$$\frac{Y(s)}{E(s)} = C(s)P(s)H(s)$$

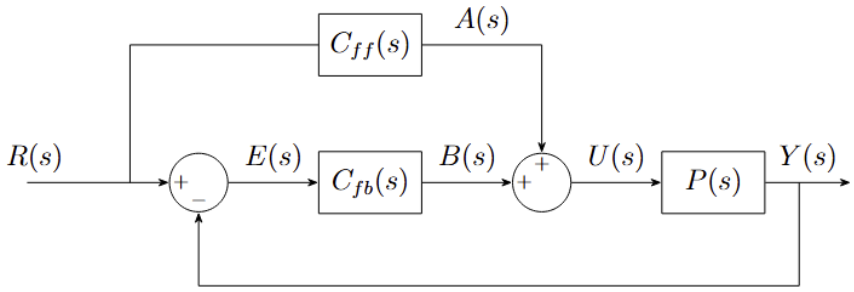
- **Closed Loop Transfer Function**
  Relates the output to the reference input

$$\frac{Y(s)}{R(s)} = \frac{C(s)P(s)}{1 + C(s)P(s)H(s)} = \frac{\text{"Forward Gain"}}{1 + \text{ "Loop Gain"}}$$

General Block Diagram Algebra

Generally, given a block diagram, we can find its transfer functions by introducing suitable signal definitions and tracing the paths.

For example:



Block diagram algebra gives the following relationships:

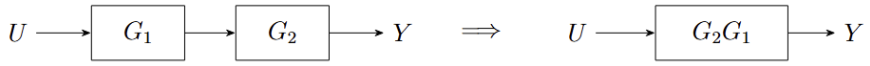
$$Y = PU, \quad U = A + B, \quad A = C_{ff}R, \quad B = C_{fb}E, \quad E = R - Y$$

Substituting these relationships into each other gives:

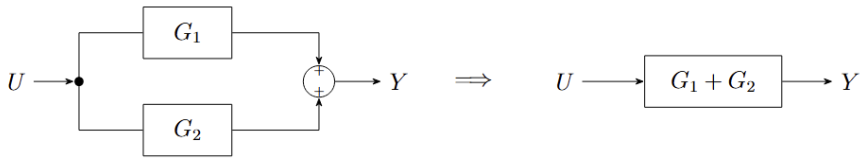
$$\frac{Y(s)}{R(s)} = \frac{P(s)[C_{ff}(s) + C_{fb}(s)]}{1 + P(s)C_{fb}(s)}$$

Equivalent Block Diagrams

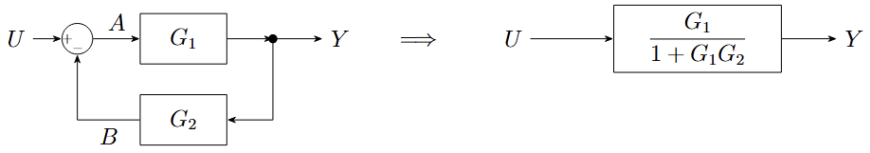
Series



Parallel

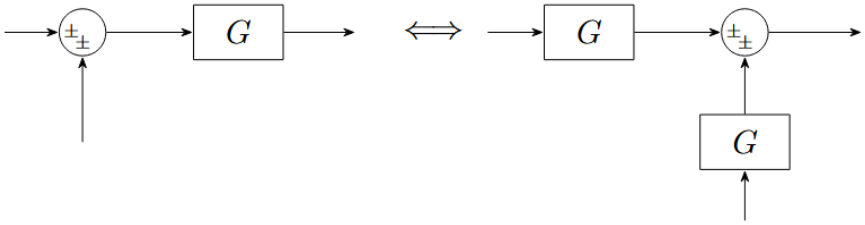


Feedback

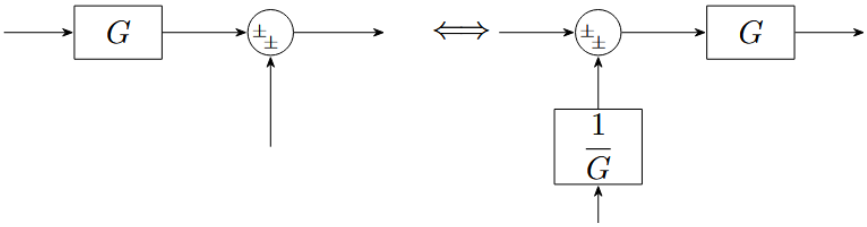


Some other (potentially) useful simplifications by moving blocks around:

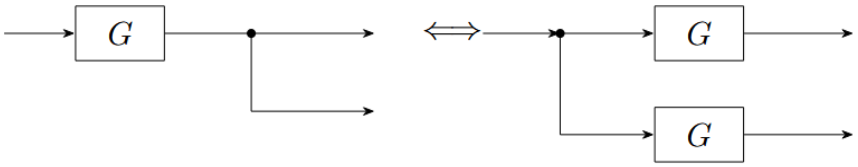
Rule 1



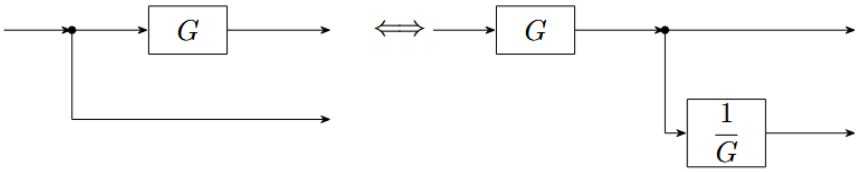
Rule 2



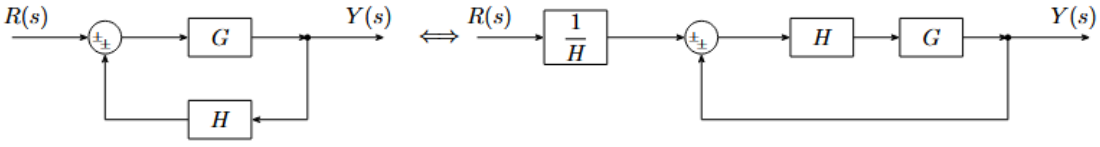
Rule 3



Rule 4



Rule 5



System Behavior and Feedback Control

DC Gain and Steady-State Error

DC Gain

The DC gain of a system is the steady-state gain of the system when subjected to a unit step input. It is derived using the final value theorem:

$$K_{DC} = \lim_{s \to 0} sY(s)$$

Given a unit step input $R(s) = \frac{1}{s}$, the DC gain can be expressed in terms of the transfer function $G(s)$:

$$K_{DC} = \lim_{s \to 0} sY(s) = \lim_{s \to 0} s \cdot G(s) \cdot \frac{1}{s} = \lim_{s \to 0} G(s)$$
$$\implies K_{DC} = G(0)$$

Note: This definition is valid only for stable systems where all the poles of $sY(s)$ are in the left half plane.

Steady-State Error

The steady-state error of a system is the difference between the desired output and the reference input as time approaches infinity. It can be derived using the final value theorem and the error transfer function $E(s)/R(s)$:

$$e_{ss} = \lim_{s \to 0} sE(s)$$

Note that the steady state error could differ depending on the type of reference input (step, ramp, etc.).

Poles and Zeros of Transfer Functions

The transfer function of any system modeled by ordinary differential equations can be expressed as a ratio of polynomials in s. Once factored, the transfer function can be expressed in terms of its poles and zeros:

$$G(s) = \frac{(s - z_1)(s - z_2) \cdots (s - z_m)}{(s - p_1)(s - p_2) \cdots (s - p_n)}$$

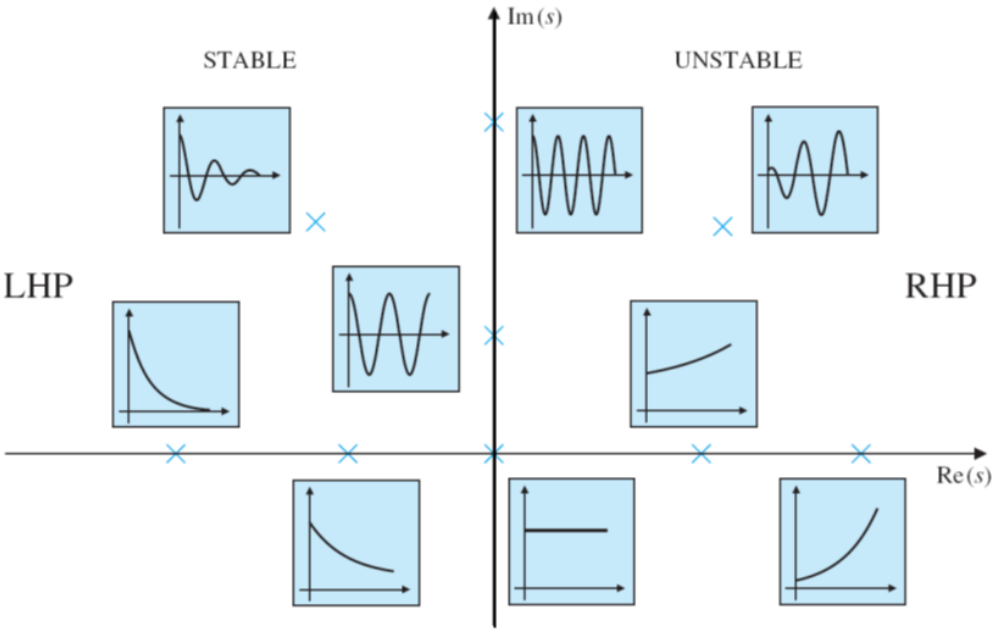
There are m zeros and n poles, where $m \leq n$:

- Zeros: $z_1, z_2, \ldots, z_m$ are the roots of the numerator.
- Poles: $p_1, p_2, \ldots, p_n$ are the roots of the denominator.

Poles provide information on the stability and frequency response of the system.

System Stability

As mentioned previously, the poles of a system provide information on the stability of the system.



- **Stable System:**
  A system is stable if its output to every bounded input is also bounded, instead of growing to infinity. This is satisfied if all the poles of the system are in the left half plane (LHP), having negative real parts $\text{Re}(p_i) < 0$.
- **Unstable System:**
  A system is unstable if its output to a bounded input grows to infinity. This is satisfied if any of the poles of the system are in the right half plane (RHP), having positive real parts $\text{Re}(p_i) > 0$.
- **Marginally Stable System:**
  A typical example of a marginally stable system is pure oscillation, where the output neither grows nor decays. This is satisfied if the system's poles are purely imaginary, having zero real parts $\text{Re}(p_i) = 0$.

System Response and Time Domain Specifications

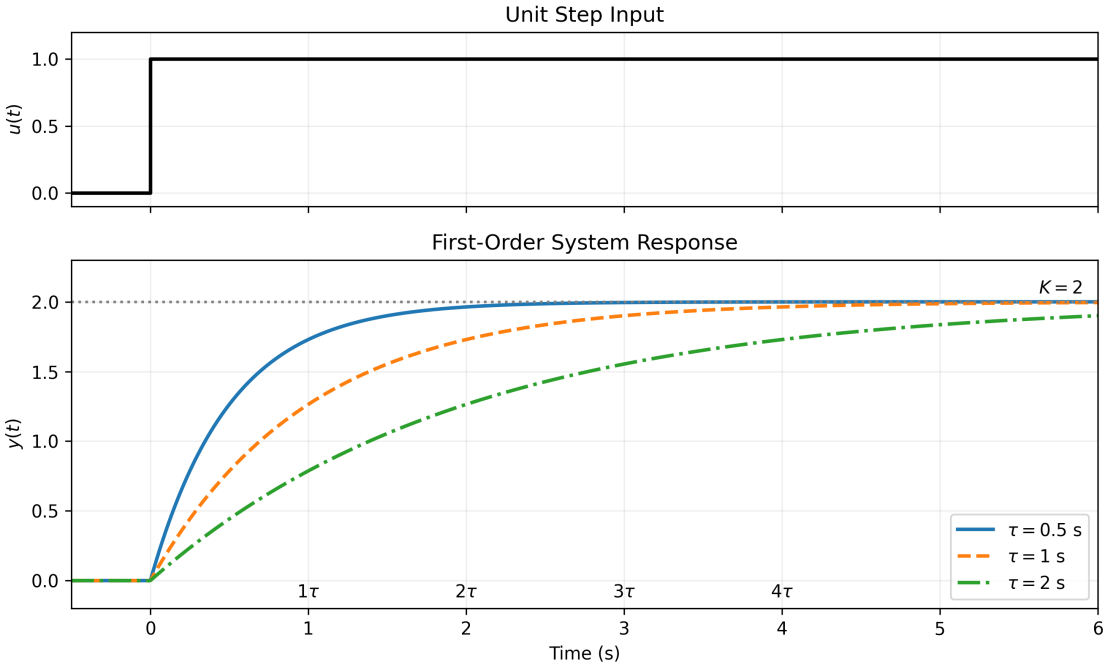
First Order System

The general characteristic equation of a first order system is given by:

$$\tau\dot{y} + y = Ku \quad \leftrightarrow \quad G(s) = \frac{Y(s)}{U(s)} = \frac{K}{\tau s + 1}$$

- **Time Constant τ**:
  Describes how quickly the system reaches steady state (shorter time constant means faster response). At 1τ, the system reaches 63.3% of its final value, at 2τ it is 86.5%, at 3τ it is 95%, and at 4τ it is 98.2%. Thus, the system is considered to have reached steady state after 4τ.

- **Steady State Gain K**:
  As defined earlier, the steady state gain here is the DC gain of the system, which describes how the output scales with respect to the input.



Second Order System

Standard Second-Order Equation

Consider the standard second-order homogenous differential equation:

$$\ddot{y} + 2\zeta\omega_n\dot{y} + \omega_n^2 y = 0$$

This describes the free response of the system. Here:

$$\omega_n = \text{ (natural frequency)}, \quad \zeta = \text{ (damping ratio)}$$

The natural frequency sets the characteristic timescale of the response, while the damping ratio determines whether the system is oscillatory and how quickly it decays.

Standard Second-Order Transfer Function

Now consider the forced response of the system with a non-zero input $bu(t)$:

$$\ddot{y} + 2\zeta\omega_n\dot{y} + \omega_n^2 y = bu(t)$$

Taking the Laplace transform of both sides, we get:

$$\left(s^2 + 2\zeta\omega_n s + \omega_n^2\right) Y(s) = bU(s)$$

Therefore, the transfer function is:

$$G(s) = \frac{Y(s)}{U(s)} = \frac{b}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

Assuming a system with a DC gain of 1, we find that $b = \omega_n^2$:

$$G(0) = \frac{b}{\omega_n^2} = 1 \quad \Longrightarrow \quad b = \omega_n^2$$

Consider the standard second-order transfer function

$$\boxed{G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}}$$

For a unit step input $U(s) = \frac{1}{s}$, the output is:

$$Y(s) = G(s)U(s) = \frac{\omega_n^2}{s(s^2 + 2\zeta\omega_n s + \omega_n^2)}$$

For the underdamped case $0 < \zeta < 1$, the poles are complex conjugates and result in an oscillatory response. Define the damped natural frequency and and phase angle as:

$$\boxed{\omega_d = \omega_n\sqrt{1-\zeta^2}, \quad \phi = \cos^{-1}(\zeta)}$$

The unit-step response in the time domain is then given by:

$$y(t) = 1 - \frac{1}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t}\sin(\omega_d t + \phi)$$

Poles, Natural Responses, and Damping Cases

The poles of the second-order system are given by setting the denominator of the transfer function (the characteristic equation) to zero:

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$$

$$\Longrightarrow \; s_{1,2} = \frac{-2\zeta\omega_n \pm \sqrt{(2\zeta\omega_n)^2 - 4\omega_n^2}}{2} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}$$

Note that the stability of the system, and the locations of the poles, depend entirely on the damping ratio ζ:

- **Undamped Case** $\zeta = 0$: Poles are purely imaginary, resulting in pure oscillation with no decay. The system is marginally stable.

  $$s_{1,2} = \pm\hat{\imath}\omega_n$$

  Where $\hat{\imath} = \sqrt{-1}$ is the imaginary number.

  The time domain response is:

  $$y(t) = C_1 e^{\hat{\imath}\omega_n t} + C_2 e^{-\hat{\imath}\omega_n t} \quad \leftrightarrow \quad \boxed{y(t) = A\cos(\omega_n t) + B\sin(\omega_n t) \quad \text{(Euler's Identity)}}$$

- **Underdamped Case** $0 < \zeta < 1$: Poles are complex conjugates with negative real parts, resulting in oscillatory decay. The system is stable.

  For $0 < \zeta < 1$, the square root term is imaginary:

  $$\sqrt{\zeta^2 - 1} = \hat{\imath}\sqrt{1 - \zeta^2}$$

  $$\implies s_{1,2} = -\zeta\omega_n \pm \hat{\imath}\omega_n\sqrt{1 - \zeta^2} = -\zeta\omega_n \pm \hat{\imath}\omega_d$$

  Where $\omega_d = \omega_n\sqrt{1 - \zeta^2}$ is the damped natural frequency.

  The time domain response is:

  $$\boxed{y(t) = e^{-\zeta\omega_n t}\left[A\cos(\omega_d t) + B\sin(\omega_d t)\right]}$$

  The system oscillates at a frequency of ω_d while decaying at in an exponential envelope of $e^{-\zeta\omega_n t}$.

- **Critically Damped Case** $\zeta = 1$: Poles are real and repeated, resulting in a non-oscillatory (exponential) response. The system is stable.

  $$s_{1,2} = -\omega_n$$

  The time domain response is:

  $$\boxed{y(t) = (C_1 + C_2 t)e^{-\omega_n t}}$$

  Recall from differential equations that the repeated root solution has to be multiplied by t to get a second linearly independent solution.

  Critical damping represents the boundary between oscillatory and non-oscillatory behavior, and is the fastest response.

- **Overdamped Case** $\zeta > 1$: Poles are real and distinct, resulting in a non-oscillatory (exponential) response. The system is stable.

  $$s_{1,2} = -\omega_n\left(\zeta \mp \sqrt{\zeta^2 - 1}\right)$$

  The time domain response is:

  $$\boxed{y(t) = C_1 e^{s_1 t} + C_2 e^{s_2 t}}$$

  The system decays exponentially with no oscillation. One pole lies closer to the origin and thus decays slower than the other (the dominant pole). Increasing damping far beyond critical damping thus slows down the response of the system.

For example: The mass-spring-damper system is a classic second-order system.

$$m\ddot{x} + b\dot{x} + kx = 0 \quad \leftrightarrow \quad \ddot{x} + \frac{b}{m}\dot{x} + \frac{k}{m}x = 0$$

Comparing to the standard form gives:

$$\omega_n = \sqrt{\frac{k}{m}}, \quad \zeta = \frac{b}{2\sqrt{km}}$$

Define the critical damping coefficient as

$$b_{\text{crit}} = 2\sqrt{km}$$

Thus, the damping ratio can be expressed as relative to the amount required for critically damped behavior (i.e. $\zeta = 1$):

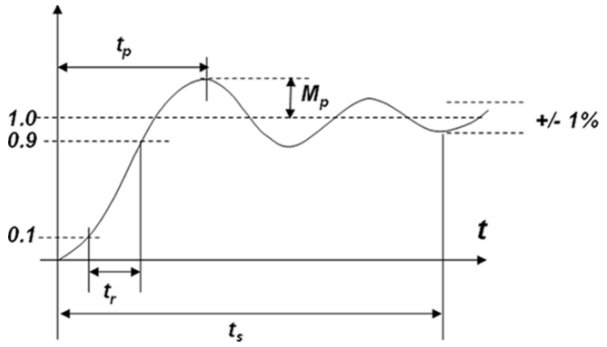
$$\zeta = \frac{b}{b_{\text{crit}}}$$

Step Response and Time-Domain Specifications

The following time domain specifications are defined for the underdamped case $0 < \zeta < 1$ under a unit step input.

Recall that the time domain response of a second-order system to a unit step input is given by:

$$y(t) = 1 - \frac{1}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t} \sin(\omega_d t + \phi)$$



- **Rise Time** t_r:
  The rise time is the time required for the response to first reach its steady state value: $y(t_r) = 1$. This occurs when the sine term is equal to 0:

  $$\sin(\omega_d t_r + \phi) = 0 \quad \Longrightarrow \quad \omega_d t_r + \phi = n\pi, \quad n = 1, 2, 3, \ldots$$

  Therefore:

  $$\boxed{t_r = \frac{\pi - \phi}{\omega_d} = \frac{\pi - \cos^{-1}(\zeta)}{\omega_n\sqrt{1-\zeta^2}} \quad \text{(0-100\% rise)}}$$

  Another common convention is to define the 10-90% rise time, which is approximately:

  $$\boxed{t_r \approx \frac{1.8}{\omega_n} \quad \text{(10-90\% rise)}}$$

  This is valid for $\zeta \approx 0.5$

- **Peak Time** t_p:
  The first peak occurs when $\dot{y}(t_p) = 0$. This gives the condition:

  $$\omega_d t_p = \pi$$

  Thus, the peak time is:

  $$\boxed{t_p = \frac{\pi}{\omega_d} = \frac{\pi}{\omega_n \sqrt{1-\zeta^2}}}$$

  Increasing the natural frequency decreases the peak time, moving the peak earlier in time.

- **Maximum Overshoot** M_p:
  The max overshoot is the amount by which the first peak exceeds the steady state value. It is defined as:

  $$M_p = \frac{y_{\text{peak}} - y_{\text{ss}}}{y_{\text{ss}}}$$

  For the standard second-order system,

  $$\boxed{M_p = e^{-(\pi\zeta)/\sqrt{1-\zeta^2}} \times 100 \quad [\%]}$$

  Notice that the max overshoot depends ONLY on the damping ratio ζ, thus an overshoot requirement immediately gives a minimum damping ratio requirement.

  $$\boxed{\zeta = \frac{-\ln(M_p)}{\sqrt{\pi^2 + [\ln(M_p)]^2}}}$$

- **Settling Time** t_s:
  The settling time is the time required for the response to be within a certain percentage of the steady state value. Recall that the responses is bounded by the exponential envelope $e^{-\zeta\omega_n t}$. Thus, define the exponential decay rate as:

  $$\boxed{\sigma = \zeta\omega_n}$$

  The settling time is then approximately:

  $$\boxed{t_s \approx \frac{4}{\zeta\omega_n} \quad (2\% \text{ criterion})}$$

  $$\boxed{t_s \approx \frac{3.6}{\zeta\omega_n} \quad (5\% \text{ criterion})}$$

Pole Locations and Performance
We can relate the time domain specifications to the pole locations of the system. But first, the poles can be graphically represented in the complex plane as:

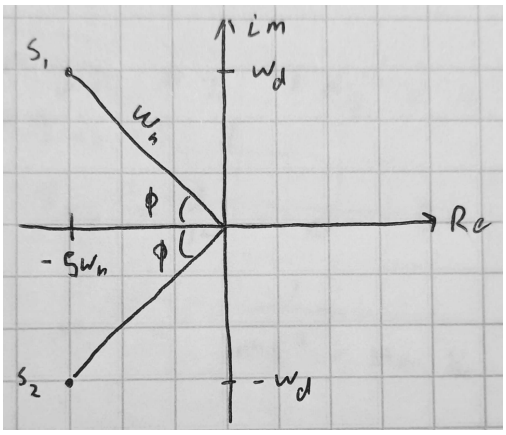
- **Magnitude**:

$$s_{1,2} = -\zeta\omega_n \pm \hat{\imath}\omega_n\sqrt{1-\zeta^2} = -\sigma \pm \hat{\imath}\omega_d$$

$$|s_1| = |s_2| = \sqrt{\sigma^2 + \omega_d^2} = \sqrt{(\zeta\omega_n)^2 + \omega_n^2(1-\zeta^2)} = \omega_n$$

- **Phase Angle**:

$$\phi = \cos^{-1}\left(\frac{\sigma}{|s_{1,2}|}\right) = \cos^{-1}\left(\frac{\zeta\omega_n}{\omega_n}\right) = \cos^{-1}(\zeta)$$

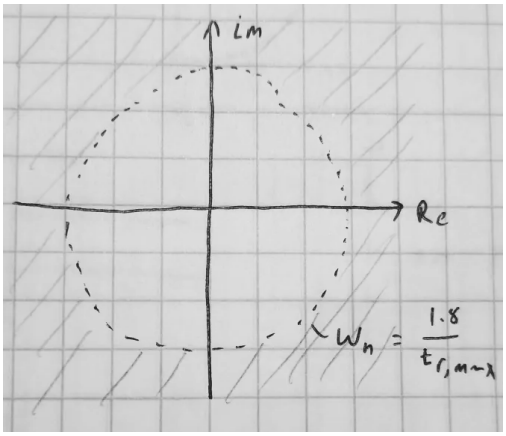


This gives a direct connection between time domain specifications and pole locations.

- **Rise Time** t_r:
  Using the 10-90% rise time approximation: $t_r \approx \frac{1.8}{\omega_n}$. Thus, for a minimum rise time requirement, we can find the minimum natural frequency:

$$t_r \leq t_{r,\max} \quad \Longrightarrow \quad \omega_n \geq \frac{1.8}{t_{r,\max}}$$

  Graphically, this corresponds to a circle of radius ω_n in the complex plane.

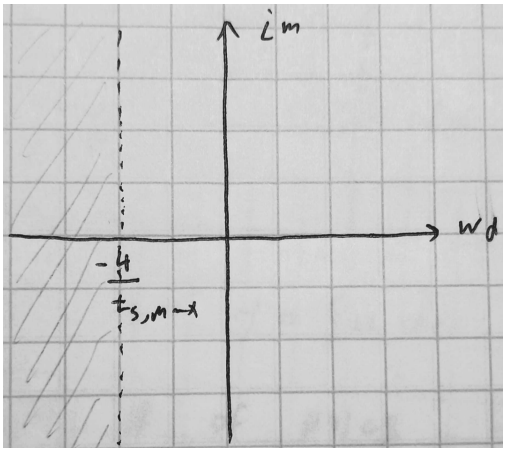


- **Settling Time** t_s:
  Using the 2% settling time approximation: $t_s \approx \frac{4}{\zeta\omega_n} = \frac{4}{\sigma}$. Thus, for a maximum allowable settling time requirement, we can find the minimum exponential decay rate required:

$$t_s \leq t_{s,\max} \quad \Longrightarrow \quad \sigma \geq \frac{4}{t_{s,\max}}$$

  Graphically, this corresponds to a vertical line in the complex s-plane at $\mathrm{Re}(s) = -\frac{4}{t_{s,\max}}$, since the real part of the pole is $\mathrm{Re}(s) = -\sigma$.

$$Re(s) \leq -\frac{4}{t_{s,\max}}$$



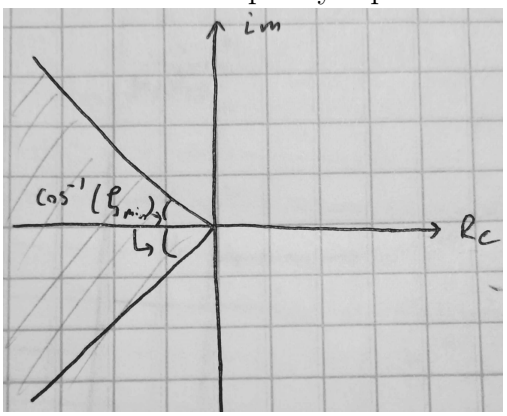
- **Maximum Overshoot** M_p:
  For a maximum allowable overshoot requirement, we can find the minimum damping ratio required:

$$\%M_p \leq \%M_{p,\max}$$

  This defines a minimum damping ratio requirement, which corresponds to a minimum phase angle in the complex plane:

$$\zeta \geq \zeta_{\min} \quad \Longrightarrow \quad \phi \leq \cos^{-1}(\zeta_{\min})$$

  Note that the inequality flips because $\cos^{-1}(x)$ is a decreasing function.

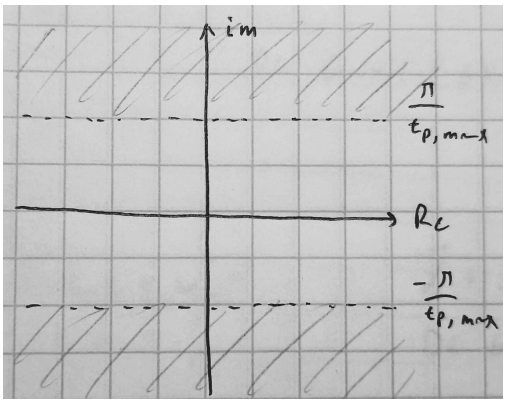


Lines of constant ζ are rays from the origin in the left half plane. Thus, an overshoot specification defines an angular region in which the poles must lie. Less overshoot corresponds to a larger damping ratio!

- **Peak Time t_p**:
  For a maximum allowable peak time requirement where $t_p = \frac{\pi}{\omega_d}$, we can find the minimum damped natural frequency required:

$$t_p \leq t_{p,\max} \quad \Longrightarrow \quad \omega_d \geq \frac{\pi}{t_{p,\max}}$$

  Since $\omega_d = |\operatorname{Im}(s)|$, this corresponds to a horizontal line in the complex plane at $\operatorname{Im}(s) = \pm\frac{\pi}{t_{p,\max}}$.



PID Controllers

PID controllers have the following general form, letting the error signal be $e(t) = r(t) - y(t)$:

$$u(t) = k_p e(t) + k_i \int_0^t e(\tau)d\tau + k_d \frac{d\,e(t)}{dt}$$

$$U(s) = (k_p + k_i \frac{1}{s} + k_d s)E(s)$$

Where the PID controller transfer function is:

$$\boxed{C(s) = k_p + k_i \frac{1}{s} + k_d s}$$

Notice that the derivative term adds a zero to the system, while the integral term adds a pole to the system. With integral control, we can eliminate steady-state error.

- k_p is the proportional gain which acts like a spring, changing how sensitive the system is to error.
- k_d is the derivative gain which acts like a damper, changing how quickly the system responds to changes in error.
- k_i is the integral gain which improves tracking, contributing to the control effort based on the accumulated error over time.

Root Locus Plots

For a negative unity feedback system, the closed-loop transfer function is given by:

$$\frac{Y(s)}{R(s)} = \frac{C(s)P(s)}{1 + C(s)P(s)}$$

The denominator of the closed-loop transfer function is called the characteristic equation:

$$1 + C(s)P(s) = 0$$

The root locus describes how the closed-loop poles of a system move in the complex plane as a single parameter (usually the gain) is varied from 0 to ∞.

Standard Evans Form

Thus, we will *factor out the gain of interest* K from the characteristic equation, giving us the standard Evans form:

$$\boxed{1 + KL_0(s) = 0, \quad K \geq 0}$$

where $L_0(s)$ is the loop transfer function without the gain K.

Because we are interested in the behavior of the closed-loop poles, instead of analyzing the full closed-loop transfer function, we can analyze the poles and zeros of the loop transfer function $L_0(s)$ to determine the locations s that satisfy the characteristic equation. These locations s exactly correspond to the closed-loop poles of the system.

Defining the loop transfer function as a ratio of polynomials, where z_j and p_i are the zeros and poles of the loop transfer function, respectively:

$$L_0(s) = k_0 \frac{\prod_{j=1}^{m}(s - z_j)}{\prod_{i=1}^{n}(s - p_i)}$$

where k_0 is the ratio of the leading coefficients of the numerator and denominator polynomials.

Rewriting the closed-loop pole condition (the characteristic equation):

$$1 + KL_0(s) = 0 \quad \leftrightarrow \quad KL_0(s) = -1$$

Putting this into magnitude and phase form gives the magnitude condition and the phase condition:

$$\boxed{|KL_0(s)| = 1 \quad \text{(magnitude condition)}}$$

For positive $K > 0$, the phase condition is:

$$\boxed{\angle L_0(s) = (2\ell + 1)\pi, \quad \ell \in \mathbb{Z} \quad \text{(phase condition)}}$$

Therefore, a point s in the complex plane is on the root locus if it satisfies the phase condition, and the corresponding gain K can be found by evaluating the loop function at that point via the magnitude condition.

$$K = \frac{1}{|L_0(s)|}$$

Evans Rules

- **Rule 0: Symmetry of the Root Locus**
  For systems with real coefficients, all complex poles and zeros occur in complex conjugate pairs. Thus,

The root locus is symmetric about the real axis

- **Rule 1: Start and End Points of the Root Locus**
  We call the poles of $L_0(s)$ the open loop poles, and the zeros of $L_0(s)$ the open loop zeros.

Branches of the root locus start at the poles of $L_0(s)$ and end at the zeros of $L_0(s)$ or at infinity.

  Letting n be the number of poles and m be the number of zeros, define relative degree as $r = n - m$. Then there are r branches of the root locus that go to infinity (asymptotes) while the remaining m branches go to the open-loop zeros.

- **Rule 2: Real Axis Segments of the Root Locus**
  Let N_R be the total number of open-loop poles AND zeros to the right of a candidate point s on the real axis. Then, a point s on the real axis lies on the root locus if

$k_0 > 0:$	N_R is odd
$k_0 < 0:$	N_R is even

  Note: We will see that while k_0 is usually positive, if it is negative, the phase condition will be slightly different (*see HW2 solutions*) which will result in a different set of real axis segments for the root locus.

- **Rule 3: Asymptotes and Centroid of the Root Locus**
  All asymptotes of the root locus intersect/originate at the centroid, which is given by:

  $$\alpha = \frac{\sum_{i=1}^{n} p_i - \sum_{j=1}^{m} z_j}{r}, \quad r = n - m$$

  The number and directions of the asymptotes are given by:

  $$\theta_q = \begin{cases} \dfrac{(2q+1)\pi}{r}, & k_0 > 0 \\ \dfrac{2q\pi}{r}, & k_0 < 0 \end{cases} \qquad q = 0, 1, \ldots, r-1$$

Frequency Response Analysis (Bode Plots)